

Tioguanine

Tioguanine is used to treat several types of acute leukaemias. It may sometimes be used to treat other cancers.

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What is tioguanine?

Tioguanine is a chemotherapy drug used to treat acute [leukaemias](#), including [acute myeloid leukaemia \(AML\)](#) and [acute lymphoblastic leukaemia \(ALL\)](#).

It may sometimes be used to treat other cancers. It is best to read this information with our general information about [chemotherapy](#) and the [type of cancer](#) you have.

Your doctor will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

How tioguanine is given

Tioguanine comes as tablets. This means you can take it at home. You might take it on its own, or with other chemotherapy drugs.

During your course of treatment, you will meet with someone from your cancer team, such as a:

- cancer doctor
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chemotherapy nurse or specialist nurse

- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

During your course of treatment, you will have regular blood tests. This is to check that it is safe for you to have chemotherapy.

You will meet with a doctor, nurse or pharmacist before you have chemotherapy. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacist team will prepare your chemotherapy.

Your course of chemotherapy

Chemotherapy is usually given as a course of several cycles of treatment over a few months. Your cancer team will discuss your treatment plan with you. They may give you a copy of the treatment plan to take home.

Your cancer team will give you the tablets to take home. They may give you tablets of different strengths. Always take them exactly as explained. This is important to make sure they work as well as possible for you. You may be given tablets of different strengths.

Your cancer team may also give you anti-sickness drugs and other medicines to take home.

Taking tioguanine tablets

Make sure you:

- do not eat for 1 hour before taking the tablets and for 1 hour after
- swallow them whole with a glass of water
- do not chew or crush them
- take them at the same time every day.

If you have trouble swallowing the tablets, you can break them in half. If you do break them, be careful not to inhale any of the tablet powder.

If you forget to take the tablets, do not take a double dose. Contact the hospital on the 24-hour number you have been given for advice.

Other things to remember about your tablets:

- Wash your hands after taking them.
- Other people should avoid direct contact with them.
- Keep them in the original package and at room temperature, away from moisture, heat and direct sunlight.
- Keep them safe, where children cannot see or reach them.
- If you are sick (vomit) just after taking them, do not take an extra dose. Take your next dose at the usual time.
- Do not throw away unused capsules or tablets. Return them to your cancer team at the hospital.

Alcohol

Your doctor may advise you not to drink alcohol while you are taking tioguanine.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Feeling sick

Your doctor, nurse or pharmacist will prescribe anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to, even if you do not feel sick. It is easier to prevent sickness than to treat it after it has started.

[If you feel sick](#), take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids. If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice. They may change your anti-sickness treatment. Let them know if you still feel sick.

Diarrhoea

This treatment may cause [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Sore mouth and throat

This treatment may cause a [sore mouth and throat](#). You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

Feeling tired

[Feeling tired](#) is a common side effect of this treatment. It is often worse towards the end of treatment and for some weeks after it ends. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

Effects on the liver

Tioguanine can cause changes in the way your liver works.

You will have regular blood tests to check how well your liver is working. Sometimes liver changes can be serious. If you have any of the following symptoms during treatment or after it finishes, you should contact the hospital straight away, or go to hospital:

- yellow skin or eyes
- feeling very sleepy
- dark urine (pee)
- unexplained bleeding or bruising
- pain in the right side of your tummy (abdomen)
- swelling in your feet or legs
- loss of appetite.

These may mean your liver is affected and you may need treatment. Your doctor may reduce the dose or stop the medication. After you stop taking tioguanine, you may still have tests to check your liver.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Tumour lysis syndrome (TLS)

Some people are at risk of developing a condition called tumour lysis syndrome (TLS). The risk is highest at the beginning of treatment. TLS happens when treatment makes large numbers of cancer cells die and break down quickly. This releases lots of waste products into the blood and can affect the kidneys and heart.

TLS can be prevented or treated. You will have regular blood tests to check for TLS. You may have:

- extra fluids through a drip
- medicines such as allopurinol tablets or [rasburicase](#) through a drip.

Drinking at least 2 litres (3½ pints) of fluid a day will also help.

Bowel problems

Rarely, tioguanine can cause a serious problem with your bowels. If you have any of these symptoms, contact the hospital straight away on the 24-hour number you have been given or go to hospital:

- severe pain in your tummy
- swelling or bloating of the tummy
- being sick (vomiting)
- diarrhoea (passing more stools than is usual for you)
- blood or mucus in your stools (poo)
- a fever, with a temperature over 37.5°C (99.5°F).

Sensitivity to sun

During treatment and for several months afterwards, you will be more sensitive to the sun. Your skin may burn more easily than usual. You can still go out in the sun, but cover up with clothing and a hat. You should use a suncream with a sun protection factor (SPF) of at least 30.

Other information

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
-

vitamins or supplements

- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Lactose intolerance

This drug contains a small amount of lactose. Lactose is a type of sugar mainly found in milk and dairy products. If you have a lactose intolerance, talk to your doctor before you start this treatment.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Sex

It is possible that small amounts of chemotherapy may be passed on through vaginal fluids or semen. If you have sex during this treatment, your cancer team will usually advise using condoms or a dental dam to protect your partner.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Changes to periods

If you have periods, these may become irregular or stop while you are having this treatment. They might return after treatment, but this does not always happen. Your [menopause](#) may start sooner than it would have done. Your doctor, nurse or pharmacist can give you more information.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

Date reviewed

Reviewed: 01 March 2025 | Next review: 01 September 2027

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help